



Emotions in Sport

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Learning Objectives

Basic:

- To be able to describe emotions
- To understand the contribution of nature and nurture for emotional expression
- To understand mind-body interaction using emotions as examples

Advanced:

- To be able to describe interindividual and intraindividual functions of emotions
- To be able to describe the influence of emotions in sport
- To be able to describe emotion regulation
- To understand emotional intelligence and its influence on athletic performance

Experts:

- To learn about methods to induce emotions and moods
- To gain knowledge about instruments measuring emotions, emotional intelligence, and mood

11.1 Introduction

Emotions play an important role in sport. The desire to experience or not to experience an emotion motivates a lot of behavior in humans. It helps explain why millions of spectators at major sporting events are spellbound in front of the television or flock to large sports arenas and live public viewing events. Think, for example, of the moment when in the USA, Michael Jordan scored the last shot in the fourth game of the 1998 National Basketball Association (NBA) final series against the Utah Jazz. This iconic moment has recently been covered in the widely viewed documentary “The Last Dance” which vividly captured the intense emotional reactions the event caused all over the world with players and spectators jumping with joy and creating a good mood among Chicago Bulls fans even days later. However, if you were a supporter of the Utah Jazz in the final, the identical situation most likely had a different effect on you as the numerous pictures of crying and mourning players and fans in the media showed. Although intense emotions like the ones described tend to be more memorable, more subtle forms of emotions are omnipresent in sports for everyone involved. For instance, athletes and spectators might be annoyed or angry when the referee decides against their own team. Or think about how you might feel when watching a super slow-motion picture of an athlete getting injured (e.g., breaking a leg). Further, athletes and fans are probably familiar with feeling nervous tension in anticipation of an important match or competition. Interestingly, most research on emotions in sports is conducted on the last example, that is, on anxiety and stress.

Therefore, this textbook contains a separate chapter on the topic of anxiety and stress in sports.

The influential emotion researcher Richard Lazarus (2000) criticizes that stress/anxiety research and research on emotions, in general, have been conducted in relative isolation from each other:

» “In the past, two almost separate research literatures have grown up, one centered on stress and coping, the other on emotion. [...] This is illogical and counterproductive.” (p. 231)

The reason that anxiety or the stress process has received a lot of attention in sport psychology has to do with two historically important publications (Martens, 1977; Martens et al., 1990a) that provided a theory and tool to examine sport-specific anxiety, and thereby providing a major thrust for research. As research evolved, it became clear these emotions are commonly experienced. It also became clear that the influence of anxiety on sports performance was complicated; some people thrive when feeling anxious, while others perform poorly (Neil & Woodman, 2019).

Reflection

Try to remember emotional experiences that you have had while doing or watching sports.
What emotions do you remember in this context?

In everyday life, we often get asked how we are feeling or how we are doing. Usually, we answer this question without much thought with relatively empty statements like “good,” “It is going well,” or “It could be worse,” without giving much detail on our complex inner world of experiencing and feeling. Although feelings and emotions seem to be so central to human psychology, there is hardly any other topic within psychology that has led to such great controversy (e.g. Beck, 2015). This is exemplified by a recent review article entitled *What Scientists Who Study Emotion Agree About* published by the prominent emotion psychologist Paul Ekman (Ekman, 2016). The controversy on emotions in psychology is also expressed by the fact that most standard textbooks in the applied field of sport psychology do not contain a chapter on emotions in sport but usually only deal with anxiety and the stress process in sport. Even though most people have an intuitive understanding of what is meant by the term “emotions” or “moods,” this understanding differs considerably from person to person (Ekman, 2016), which complicates both research and applied interventions targeting emotions. Emotions in everyday life are usually thought to be merely a feeling (e.g., “I am happy” or “I am angry with my colleague”). However, common psychological theories define emotions more broadly.

Emotions

Emotions describe a complex pattern of physical and mental changes in response to a triggering situation perceived as personally significant. These changes include physiological arousal, feelings, cognitive processes, expression, and behavior. We are usually emotional about something or someone, and these feelings are intense and relatively short.

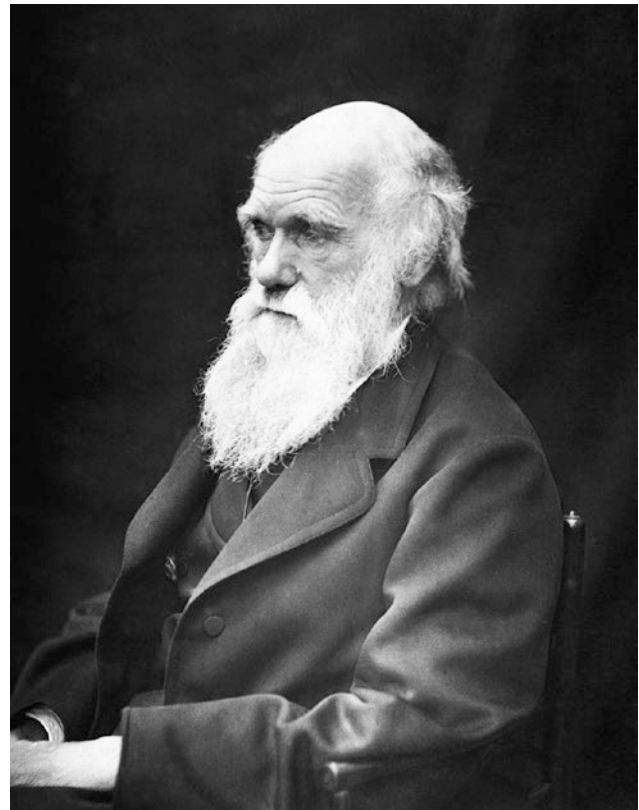
Moods

Moods are distinguished from emotions primarily by the fact that emotions are considered to be reactions to specific events that are short term and intense. Moods are usually less intense, do not have a specific antecedent or cause that an individual can identify, and can last for several hours or days. There is a weaker relationship between moods and their triggering events.

11.2 Universal Emotions and Culture: Darwin's Observations on the Connection Between Emotions and Evolution

When we travel to foreign countries and cultures, communication is impeded by our relative inability to master the different languages. In contrast, Charles Darwin (■ Fig. 11.1) noted that the experience and expression of emotional states seems to be relatively universal and cross-culturally recognized, thereby facilitating intercultural communication beyond language. In his book *The Expression of the Emotions in Man and Animals*, Darwin (1872/1998), describes that emotions have developed hand in hand with other important structures and functions of living beings. Darwin regarded emotions as adaptive mechanisms representing highly specific, coordinating operations of the brain. These operations had the purpose of preparing a living being for certain categories of reoccurring situations/stimulus patterns in the environment and in the body. They also help them to respond adaptively to recurring problems and opportunities in their environment (e.g., threat, disgust, or feelings of sexual arousal). Darwin saw emotions as inherent, specialized physiological programs that increase the chances of living beings to pass on their genes to the next generation. Or as de Waal (2019) puts it, “emotions help us navigate a complex world that we don’t fully comprehend.”

According to Darwin's theoretical considerations, physiological processes and muscle movements associated with emotions generally fulfill two functions: 1) preparing the organism for adaptive actions in reoc-



■ Fig. 11.1 Charles Darwin

curing stimulus constellations and 2) communicating important social cues.

? How can the main ideas of Darwin's “theory of emotional expression” be summarized?

- Emotional expressions developed hand in hand with other physiological reactions to prepare an organism for adaptive actions (e.g. fight or flight). These emotional expressions also signal to others that adaptive actions are likely.
- If emotional signals are useful to the individual (i.e., increased reproductive fitness), the communicative function might become more important than the physiological function. In turn, evolutionary selection pressure would lead to exaggerated expressions, and physiological functions might become vestiges. As a consequence, contrary messages (e.g., pride and shame) are often expressed by opposing movements and postures to make them better recognizable by observers (e.g., pride takes up more place in space, whereas the expression of shame makes you collapse and take up less place in space; principle of antithesis).

A recent theory has taken up these considerations and integrated them into the “two-step model of emotional expression” (see for an overview Shariff & Tracy, 2011). In this two-step model, it is suggested that internal phys-

iological regulation drives the intensity of emotional expressions (e.g., dilated pupils in anxiety to identify the source of a possible threat). Later in the process, however, social communication is more important, since the efficient organization of groups leads to a range of adaptational advantages (e.g., signaling to the social group through dilated pupils that danger was imminent). Another example illustrative of Darwin's theoretical considerations is the evolution of display behaviors and threatening gestures. Behaviors such as staring at an

enemy, flaring and standing up, tensing muscles, and displaying one's weapons were used to prepare for a fight. All these behaviors could be interpreted by an observer or rival as signs of imminent aggression and danger. The signaling and interpretation of aggressive or dominant behavior gave both parties a survival advantage, as the probability of a life-threatening fight decreased (► [Study Box: "This Is What Winners Look Like": Can Spectators Judge the Score Based on Body Language?](#); ► [Side Story: Body Language and Emotions in Sport](#)).

Case Study Box

"The Expression of Winning": Can Spectators Judge the Score Based on Body Language?

From Darwin's theory on the evolution of display behavior follows that people in confrontational situations should display nonverbal behavior that provides information about who is superior or inferior (see Furley, 2021, for an evolutionary account of emotions and applications in the sport context). Furley and Schweizer (2014a, 2014b) based a study on this hypothesis, examining it in athletic competitions and whether people can distinguish between leading and trailing athletes based on their nonverbal behavior.

In Study 1, participants viewed short video recordings of basketball

and table tennis players playing in competition. The actual score at the time of the recordings varied from "trailing by far" to "leading by far" for the player shown. For each video recording, participants assessed how far the athletes were leading or trailing on a continuous scale.

Study 2 investigated developmental differences by comparing the results of Study 1 with one group of children aged 4–8 years and a second group of children aged 9–12 years.

Study 3 served to replicate Study 1 with new stimulus material (recordings of handball players) and investigate if domain-specific sports knowledge moderated the effect. Participants were either experienced

handball players or handball laymen or laywomen.

The results of these three studies showed that participants were able to distinguish between leading and trailing athletes. The effect size was consistently high by statistical convention. Interestingly, participants with sport-specific experience were not more accurate in their assessments, and so knowledge of the sport and its culture were not influential. Although children were also able to distinguish between leading and trailing athletes, the effect was more pronounced in adults.

Side Story

Body Language and Emotions in Sports

There is hardly a television broadcast in sports in which the commentator does not mention the body language of the athletes (see Furley & Schweizer, 2020 for a recent review on body language in sports). Statements such as "You can see the team has given up" or "The player is bursting with self-confidence" are commonplace in sports like football.

Charles Darwin observed and documented in detail that certain emotional states are communicated uniformly. Therefore, it is not surprising that we are so familiar with

body language and can interpret it reliably. After all, language developed late in evolutionary history, and our ancestors were dependent on nonverbal communication (see Fridlund, 1994, for an alternative explanatory approach, arguing that nonverbal communication has the adaptive advantage of communicating social motives and behavioral intentions rather than just emotions).

A particularly important category of nonverbal behavior, especially among primates, is the communication of dominance and submissiveness to quickly and efficiently convey information about rank and status

(Furley, 2019) and, thus, prevent possible life-threatening attacks. In this context, studies have made the remarkable observation that winning or losing in competition leads to changes in blood hormone levels (Booth et al., 1989; Edwards & Casto, 2018) associated with dominant or submissive behavior. Against this background, the frequent comments on body language in football also make sense. Recent studies show that individuals are extremely efficient at assessing the correct score tendency based solely on nonverbal signals (Furley & Schweizer, 2016) and that even children with autism spectrum

disorder are capable of doing so (Ryan et al., 2016).

Of greater practical relevance are studies showing that dominant and submissive nonverbal behavior have an important influence on the opponent (Furley & Dicks, 2012; Furley et al., 2012a). The experiments demonstrate that a dominant body language is unconsciously associated with positive characteristics such as self-confidence and competence. Conversely, a submissive body language is associated with negative characteristics such as insecurity or

fearfulness. Even the slightest signals of dominance or submissiveness activate stored schemata. For example, submissive body language leads opponents to assess them negatively and then to strengthen their perception of being able to counter them successfully. Numerous studies on self-efficacy show that the belief to perform successfully is closely related to performance (Budden et al., 2020; Feltz & Öncü, 2014). Conversely, doubts about success often lead to failure of the action (Furley & Schweizer, 2014a). Theoretically,

it would, therefore, be conceivable that an athlete's nonverbal behavior could be part of a vicious (or virtuous) circle. For example, a team trailing in a competition might, in turn, show submissive body language. This is perceived by the opponent who in return grows in self-confidence and so is more convinced in his ability to beat the opponent. This might result in the opponent actually playing more successfully and again displaying more dominant body language.

Sport Practice

How Do I Hide My Nervousness During the Penalty Kick and Increase My Chances of Success?

Two studies provide some insight into this question by showing a connection between emotions and nonverbal behavior (Jordet & Hartman, 2008; Jordet, 2009). These studies show that football players tend to display certain nonverbal avoidance behaviors during very important penalties. More specifically, they tend to turn their backs to the goalkeeper and spend less time between the referee's whistle and the start of their run-up, possibly because they want to get the important penalty kick over as quickly as possible or it could be that the players are hiding from the stressors, albeit nonconsciously. Such behavior has been shown to have a negative influence on the shooter's probability of success. It is argued that this shows that the emotion fear/anxiety is associated with poorer performance (► Chap. 12). It could also partly be due to the extent to which the goalkeeper has learned from observing such preperformance nonverbal behavior. Research has shown that goalkeepers feel more confident in saving the penalty kick when they perceive the striker is uncertain on where to place the shoot and believe they can adapt their behavior accordingly—i.e., by waiting longer until they initiate their movement (Furley et al., 2012b). Interestingly, it is precisely those nations showing this avoidance behavior most frequently that have been less successful in penalty shootouts in the past (e.g., England!!!!).

Taken together, evidence-based recommendations for penalty takers could be as follows: (1) show high self-confidence by how you pick up the ball, (2) estab-

lish good eye contact with the goalkeeper, (3) place the ball on the spot, (4) look at the goalkeeper when running back to the starting point, (5) take a deep breath after the referee's whistle and allow yourself 2 s (possibly count “21, 22”), and (6) initiate the run-up.

A straightforward hypothesis that can be derived from Darwin's theoretical considerations is that certain emotional reactions should occur universally irrespective of cultural origin of a person; emotions are universal and it was the same adaptive processes that shaped them over generations of evolution. Various researchers have tested this universality hypothesis by comparing the emotional responses of newborns and the consistency of facial expressions across cultures. If Darwin's theory was supportable, children all over the world should show comparable emotional responses (Izard, 1994). The first psychologist to provide evidence for this assumption was Silvan Tomkins (1962, 1981). He was able to show comparable, unlearned emotional responses in infants when loud sounds elicited fear. These results were interpreted as a kind of neural “hardware program” in response to a certain kind of triggering stimulus patterns. Certain environmental stimuli lead to neuronal response patterns that are automatically interconnected to brain areas and effector organs leading to an emotional response that is universal enough to fit a wide range of situational conditions. Comparative cultural research confirms this theorizing by providing evidence that some emotional responses in young children are very similar (Camras, 1992; Camras et al., 1992; ► Case Study Box: Do Infants Show Comparable Emotional Reactions?).

Case Study Box

Do Infants Show Comparable Emotional Reactions?

Five and 12-month-old infants from the USA and Japan reacted very similarly when the experimenter grasped the children's wrists and fixed them to their stomachs. The children's reactions were all recorded on video and compared with each other using a

coding system. Both the children's facial expressions and behavior were highly comparable between Japanese and American children (Camras et al., 1992).

However, this finding was not supported by a later study led by the same author (Camras et al., 1998). In this study, 11-month-old infants from

China reported less intense emotional expressions than their Japanese and the American peers. This finding could be interpreted that culture might be having an early influence on the emotional hardware program of children, and clearly further evidence is needed.

Various studies on **universal expressions** of emotions suggest that the interpretation and recognition of emotional expressions has also an evolutionary innate component. For example, 4- to 6-month-old infants could reliably distinguish between surprise, fear, and anger (Serrano et al., 1992). In addition, infants show more approach behavior and smile more often in response to looking at happy facial expressions. Infants show avoidance behavior and frowning in response to looking at angry facial expressions (Serrano et al., 1995). Infants, therefore, seem to be able to distinguish not only basic emotional expressions but also show an early understanding of their meaning.

Probably the most prominent proponent of this **basic emotion approach** is Paul Ekman, who was the only psychologist to be named as 1 of the 100 most influential people by Time Magazine in 2009. In his detailed intercultural studies, Ekman (1984, 1992, 1994) has shown what Darwin first suspected—that is, the existence of a kind of universal emotional language (at least for some facial expressions), which presumably is an innate component of the human evolutionary heritage.

? Thanks to Paul Ekman and his work on facial expression, emotion, and deception, we have a better understanding of how the expressions and gestures we display on the surface are a direct reflection of what is going on in the neurocircuitry deep inside our brains (Quote from the laudation of the Time Magazine from 30th April, 2009).

Specifically, it has been demonstrated that the emotional facial expressions of joy, anger, disgust, fear, and sadness (and possibly also surprise and contempt, depending on the study) are shown and reliably recognized all over the world (Ekman & Friesen, 1986, ■ Fig. 11.2). Regarding basic emotions, it is important to mention that not all emotional expressions are universally recognized. This has only been shown for the five emotions: joy, anger, disgust, fear, and sadness (although there is

still debate here). According to Ekman (2007), all other facial expressions are composed of a mixture of these primary facial expressions (see Gendron et al., 2018, for a recent critique of this theory).

? Based on his many years of research on emotional expressions, Paul Ekman and his daughter Eve Ekman have launched the online Atlas of Emotions in 2016 (► <http://atlasofemotions.org/>). The goal of this interactive tool is to sensitize and improve the emotional awareness of people. This, in turn, aims to help people recognize how emotions affect their lives. The atlas is based on the “neurocultural theory of emotion” (Ekman, 1994) and assumes there are (at least) five universal emotions all people can recognize and express, regardless of their origin or social background. The Atlas of Emotions is summarized in ■ Table 11.1.

? What does it mean to assume emotions as basic? The adjective basic contains two main characteristics regarding emotions (Ekman & Cordaro, 2011): One is that these emotions are discrete, meaning they can be clearly distinguished from other emotions. Differentiation criteria include facial expressions, the voice, the physiology of the autonomic nervous system, and the triggers initiating the emotion.

Second, emotions are an evolutionary product of our adaptation to the environment. Although humans are incredibly versatile in their behavior, they embody biological mechanisms that allow them to deal with basic life tasks (e.g., loss, frustration, success). Every basic (universal) emotion guides human behavior in a certain direction that has proven to be superior to other directions in the course of evolution (see Gendron et al., 2018, for a critique of this view).

In regard to the popular nature vs. nurture debate, it should be noted that the basic emotion approach does not argue that emotions are 100% nature and 0% nurture.



■ **Fig. 11.2** Evaluation of emotional facial expressions (Author: Icerko L  dia). Assign these seven emotional terms to the faces shown: fear, disgust, joy, surprise, contempt, anger, and sadness

■ **Table 11.1** Ekman's Atlas of Emotions

Emo- tion	Experience	Condition (sorted by intensity)	Response	Trigger	Moods
Anger	We get angry when something blocks us or when we think we're being treated unfairly	Annoyance Frustration Exasperation Argumenta- tiveness Bitterness Vengefulness Fury	Dispute Be passive- aggressive Insult Quarrel Scream/yell Simmer/ brood Suppress Use physical force Undermine	Interference with locomotion Interference with action Rejection by a loved one Inefficiency or bureaucracy Being put down by an authority figure Encountering Offensive beliefs Being wrongfully accused	Irritable: predisposed to becoming angry, easily provoked

(continued)

Table 11.1 (continued)

Emotion	Experience	Condition (sorted by intensity)	Response	Trigger	Moods
Enjoyment	Enjoyment describes the many good feelings that arise from experiences both novel and familiar	Sensory pleasure Rejoicing Compassion/joy Amusement Schadenfreude Relief Peace Fiero Pride Naches Wonder Excitement Ecstasy	Exclaim Engage/connect Gloat Indulge Maintain Savor Seek more	Sensory experience of nature Helping others Social interaction Playing sports Places associated with enjoyable memories The taste of something nice Spending time with loved ones	Elated: a long-lasting, generalized good feeling
Disgust	Feeling disgusted by what is toxic helps us to avoid being poisoned, physically or socially	Dislike Aversion Distaste Repugnance Revulsion Abhorrence Loathing	Avoid Dehumanize Vomit Withdraw	Anything coming out of the body Rotting or decay Fans of opposing sport teams Unfamiliar religious customs Eating insects or raw meat	Sour: generally repulsed
Fear	Our fear of danger lets us anticipate threats to our safety and, thus, ensures security and well-being	Trepidation Nervousness Anxiety Dread Desperation Panic Horror Terror	Avoid Freeze Hesitate Ruminate Scream/yell Withdraw Worry	Imminent bodily impact Sudden loss of gravity Threat to safety snake-like shapes Threat of losing job Thunder Public speaking	Apprehensive: anxious that something bad will happen, on edge
Sadness	Sadness is a response to loss, and feeling sad allows us to take a timeout and show others that we need support	Disappointment Discouragement Distraughtness Resignation Helplessness Hopelessness Misery Despair Grief Sorrow Anguish	Feel ashamed Mourn Protest Ruminate Seek comfort Withdrawal	Being rejected by someone important Losing a loved one Perceiving a loss of status Not being invited to a party Losing a treasured belonging	Dysphoric (or feeling blue): an enduring feeling of discouragement or disappointment

The interactive Atlas of Emotions with more detailed descriptions can be found at ► <http://atlasofemotions.org/>

Ekman (1994) claims in his “**neurocultural theory of emotion**” that both evolution (through the pre-connection of neural circuits in the brain) and culture influence emotional programs and their expression.

The influence of culture on emotional programs occurs via social learning and socialization, which teaches the appropriate handling of emotion according to the social situation. In this way, social rules or cultural norms are established about when and to what extent certain emotions are expressed or lived out (Mesquita & Frijda, 1992; Ratner, 2000).

The zoologist Ernst Mayr’s (1974) distinction between open and closed genetic programs is helpful in understanding the relative influences of nature and nurture on emotions. In a closed genetic program, no “add-ons” through experience are possible, whereas open genetic programs can integrate additional input through experience. Mayr argues that organisms with long developmental periods of parental care have more time to learn and adapt to varying environmental conditions and therefore have a selection advantage through open genetic programs (see Sterelny, 2012, for a detailed argumentation on this). In this context, Ekman and Cordaro (2011) argue emotions belong to the open genetic developmental programs since humans need to learn to adapt their universal emotional programs to their living conditions through experience.

According to this view, universal emotion programs evolve in the process of ontogenesis by the interconnection of sensory cells, neuron groups and effector organs (e.g. muscles), but can be influenced by experiential input.

Reflection

Am I Laughing Because I’m Happy, or Am I Happy Because I’m Laughing?

Research suggests that there is evidence for both and that our facial expressions influence our emotional experience (■ Fig. 11.3, Adelman & Zajonc, 1989). To test this so-called facial feedback hypothesis (Strack et al., 1988), participants were instructed to show different facial expressions while viewing a series of photographic slides (Rutledge & Hupka, 1985). The results showed that participants reported that the slides made them happier or less annoyed when they made a happy facial expression compared to an angry expression.

Try it yourself: Pull your eyebrows down and together at the same time. Lift your upper eyelids, tighten your lower eyelids, make your lips narrow, and press them together. If you hold this expression for a while and notice you feel slightly upset, you have just experienced the effect of facial feedback.



■ Fig. 11.3 Laughing/smiling can lift your mood

The “Try it yourself example” on facial feedback shows that we can control our facial expressions at will and the human facial expression is not just a display that shows others what evolutionary emotional program is currently running. Thus, humans can consciously put on facial expressions and suppress automatic, “real” facial expressions, as predicted by **dual-process models** (Furley et al., 2015b, for an overview in sports). For example, you can put on a fake smile to hide a lie. However, some cues allow for distinguishing real expressions from fake ones (Ekman, 2007). Ekman was able to show that fake emotions are often interrupted by very short facial expressions (**micro-expressions**) of the true emotion. Such micro-expressions usually last about 0.05 s, but with appropriate training, they can be recognized without technical aids.

Besides micro-expressions, other cues allow to distinguish between real and staged expressions. In 1862 the French anatomist Duchenne claimed a real smile could be distinguished from a fake smile by looking at the two muscles contracted during a real smile: the *musculus orbicularis oculi*, which surrounds the eyes, and the *musculus zygomaticus major*, which lifts the corners of the mouth. According to Duchenne, people do not have any problems to arbitrarily contract the *zygomaticus major* muscle to lift the lips, whereas most people cannot contract the *orbicularis oculi* muscle purposefully. With a real smile, both eyes and mouth laugh, whereas with a fake smile, usually only the mouth laughs. Therefore, the real smile was called a **Duchenne smile** (Ekman & Davidson, 1993).

? Universal emotions and culture

- Universal emotions serve a specific purpose and help people deal with a range of reoccurring situations.
- Universal emotions vary in the intensity by which they are experienced.
- We react to our universal emotions with certain actions or action tendencies.

- A trigger is something in our environment and in our thoughts that provokes an emotion.
- Nature and nurture influence whether and to what extent basic emotions become noticeable in expression or behavior.
- Emotions and expressions are in a reciprocal relationship with each other (facial feedback): Emotions influence expressions and expressions influence emotions.
- If certain basic emotions occur several times in a certain period or over a longer time, they influence our mood.

So far, research has been covered showing that some physiological responses to emotional events have a genetic origin but can be influenced by social learning. Based on these findings, we now turn to emotion theories explicitly dealing with the interaction of physiological and psychological processes.

11.3 Physiology of Emotions

Everyone knows the feeling of experiencing a strong emotion: For example, you notice your heart beats faster, your breathing becomes faster, your muscles become tense, and your mouth begins to dry. Many further physiological processes can accompany emotional reactions that do not necessarily have to be consciously noticed. All these physiological reactions evolved to mobilize the organism to respond adaptively to the current environmental conditions (Kalat, 2019).

In this process, the **autonomic nervous system** plays a key role in preparing the body for adaptive reactions by orchestrating the activity of the parasympathetic and sympathetic components. The activity ratio of the respective parasympathetic and sympathetic systems depends on the quality and intensity of the triggering stimuli.

In the present chapter, we give a schematic overview of the most important general physiological principles of emotions rather than enter into the highly complex discussion on the physiological processes of emotions in detail. It is generally accepted that the sympathetic nervous system directs its activating effect by releasing the hormones adrenaline and noradrenaline from the adrenal gland and thus controls a cascade of other physiological processes, which, for example, stimulates other internal organs, increases blood sugar, or increases sweat production. In contrast, the parasympathetic system is considered to have a deactivating or calming role, for example, by inhibiting the release of activating hormones into the bloodstream. Comparative cultural studies suggest that the autonomic nervous system is an

important component of our innate emotional hardware preparing people for a wide range of reoccurring emotional situations (Levenson et al., 1992 for more detail). Research results have also shown that participating in a sport competition have a measurable impact on hormone levels in athletes (Edwards & Casto, 2018).

In addition to the autonomic nervous system, the central nervous system needs to be briefly discussed to enable a comprehensive understanding of the physiological hardware of emotions. The **hypothalamus** and the **limbic system** are responsible for the interaction of hormonal and neural aspects of emotions. At the center of neuronal emotion research is the **amygdala**, which is part of the limbic system (Pinel & Barnes, 2018). Metaphorically, the amygdala is often described as the “gate of emotions” and as a kind of “filter for memory.” In this sense, the amygdala “colors” information from the environment and memory in a way that gives it emotional meaning or valence (i.e., pleasure or displeasure). This process metaphorically describes the mechanism by which experience influences the open basic emotional programs of humans.

The cortex, the seat of many memories, has also been shown to be involved in emotional experiences. Metaphorically speaking, sensory impressions are emotionally colored and stored as the foundation for future behavior. This process points to a close interconnection of cognition and emotion, which is also increasingly anchored in the literature (e.g. Damasio, 1994: “Somatic Marker Hypothesis”; Slovic et al., 2007: affect heuristics). In his *Somatic Marker Hypothesis*, Damasio argues that perceptual events—imagined or generated by one’s actions—are related to certain somatic reactions of the body and are stored in memory together with these somatic/emotional reactions. This means that recalling certain memories automatically activates somatic representations that were active when experiencing (or encoding) the memorized event. This gives perceptual events and anticipated perceptual events a certain valence, which is transmitted via emotional reactions (for Seligman et al., 2016, the emotional coloring of imagined prospective actions could even be the central role of emotions since the direct influence of emotions on behavior is often difficult to prove).

As a result, when making decisions and planning our behavior, we can draw on emotional valence information and thus adapt our decision-making and anticipation behavior effectively to the environment. In other words, people seem to attribute emotional valence to different action alternatives, which help us to choose between these alternatives. People who lose this function due to brain lesions have great problems to regulate themselves in everyday life, as shown in the case of Phineas Gage (► [Side Story: The Case of Phineas Gage and How It Changed Previous Thinking About Emotions](#)).

Side Story

The Case of Phineas Gage and How It Changed Previous Thinking About Emotions

The young railway worker Phineas Gage was the victim of a tragic industrial accident in 1848. During a blowup, he was shot through his face, skull, and brain with a 3-cm-thick, 90-cm-long ramming iron. This caused a lesion in the orbitofrontal and prefrontal cortex. Miraculously, Gage survived the accident. However, he appeared to be a different person thereafter. Previously, Gage

was described as a responsible, intelligent, and social person and was very popular with friends and colleagues. After he recovered from his accident, Gage appeared to be as intellectually and physically capable as before, but both his personality and his emotional world had changed completely. Above all, his disrespect, vulgarity, and impulsiveness were noticeable. As a result, he lost his job and was no longer able to take on a position involving responsibility. He became a vagabond who wandered the coun-

try in the San Francisco area until his death. Although Gage was buried along with the ramming iron that had injured him, neurologist John Harlow received permission to examine Gage's body and the ramming iron 5 years after the funeral. Today, the skull and the ramming iron are exhibited in a museum at Harvard University, and Gage's condition is now called frontal brain syndrome since the frontal brain performs important functions in terms of self-control, impulse control, and executive functions.

The cortex can thus be seen as a kind of archive in which associations are stored that connect psychological experience with physical reactions. In this context, some researchers have suggested that emotions lead to two distinct behavioral tendencies: **approach and avoidance behavior** or **approach and avoidance motivation** (Davidson et al., 2000). In this context, Davidson et al. (2000) speak of two different perceptual-action systems that are governed by emotions and are responsible for approach- and avoidance-related reactions.

➤ Physiology of Emotions

- The autonomic nervous system prepares the body for emotional reactions using the parasympathetic and sympathetic components in order to adapt.
- The hypothalamus and the limbic system are responsible for hormonal and neuronal interactions of emotion.
- Via the limbic system/amygdala, incoming information and memory content receive emotional valence.
- Emotional valence stored in networks involving the cortex influences information processing and behavior (*somatic marker hypothesis*, affect heuristics).
- Basically, a distinction can be made between an approach-related and an avoidance-related emotional system.

11.4 The Connection Between the Physiology and the Psychology of Emotions

The previous section summarized evidence that the human organism, through evolution and learning, shows a variety of rapid physiological reactions to reoccurring, meaningful events, called emotions. All these rapid reactions serve the purpose of enabling adaptive behavior. In the following, we will briefly present basic theories on the origin of emotions integrating the interaction of physiological responses, cognitive evaluations, and behavioral tendencies.

11.4.1 James-Lange Theory

Aristotle already assumed that a psychological feeling sets in after the body has reacted to a situation. The founding father of American psychology, William James, also assumed this sequence: physiological reaction followed by psychological emotion (James, 1890). This view of emotions coming from physical feedback has become known in the psychological literature as the “**James-Lange theory of emotion**” (Carl Lange was a Danish psychologist who formulated a similar theory at the same time). According to this theory, the perception of a situation leads to somatic reactions, which in turn leads to the experience of a psychological emotion.

11.4.2 Cannon-Bard Theory

The American physiologist Walter Cannon (as well as another physiologist: Philip Bard) rejected the theory of “James Lange” due to empirical evidence not being consistent with the theory (Leventhal, 1980, for an overview). Furthermore, it had been argued that the reactions of the autonomic nervous system are too slow to be the cause of the rapidly developing psychological emotions. According to Cannon, the thalamus registers incoming information, which is then transmitted in parallel to different areas. They simultaneously cause emotional experience and further somatic reactions. According to this theory, an emotional stimulus is at the same time the trigger for physical arousal and subjective experience of the emotion. A central hypothesis of this theory implies that physical arousal and emotional experience are not mutually dependent. However, more recent cognitive assessment theories suggest that this does not seem to be true.

11.4.3 Appraisal Theory (Lazarus-Schachter Theory)

In many situations, internal states of arousal are very similar, even when emotions are different. Therefore, there is a risk of ambiguity and confusion in the physiological excitation pattern. For this reason, Schachter and Singer (1962) postulated that the psychological experience of emotions is a joint effect of physiological arousal and cognitive assessment. Physiological arousal is only the first step in the emotional chain. This physical arousal is assessed (typically unconsciously) by the person in terms of its significance. Another supporter of the central role of cognitive assessment in the emergence of emotions was Richard Lazarus (Lazarus, 1991; Lazarus & Lazarus, 1994). He argued that emotions cannot be fully understood through physiological processes, but rather they result from the dynamic interaction of the person with the environment, which is continually being evaluated nonconsciously and consciously.

? Are there discrete emotions or do emotions differ only in terms of a few dimensions?

- A lengthy controversy within emotion research concerns the question of whether there are different discrete emotions (Ekman & Cordaro, 2011, for

an overview) or whether emotions are better characterized as a two-dimensional (or multidimensional) construct that varies on the dimensions valence (pleasant vs. unpleasant) and activation (high vs. low) (e.g., Russell et al., 1989).

- This controversy continues and cannot be clearly resolved in favor of one or the other perspective at this time.
- Without going too deeply into this debate (see Gendron et al., 2018, for an up-to-date overview), both the supporters of the discrete emotion approach (e.g. Ekman, 2016; Lazarus, 2000) and the supporters of the dimensional approach (Crivelli & Fridlund, 2018, for an up-to-date overview) provide conclusive arguments in favor of the respective theory.
- First studies (e.g., Cowen & Keltner, 2017) have meanwhile made promising progress in integrating the two theoretical approaches.

Even though there is plenty of empirical evidence for the important role of cognitive evaluation processes in the experience of emotions (Lazarus, 1991, 2000, for an overview), this theory is not without criticism (e.g. Zajonc, 2000), since studies suggest certain emotions cannot result from evaluation processes. Taken altogether, however, it can be said that cognitive evaluations can make an important contribution to emotional experience but are not the only source (Izard, 1993). In some situations, emotional experience appears to be controlled of innate, evolutionary “programming” of the human nervous system occurring independently of cognitive assessment. These different sources of emotional experience might be indicative that emotions fulfill different functions in humans. The assumed functions of emotions will be discussed in the next section.

> Psychology of Emotions

- **James Lange Theory:** Experiencing emotions arises from physical/somatic feedback.
- **Cannon-Bard Theory:** Incoming information is registered by the thalamus and simultaneously influences physical arousal and emotional experience.
- **Lazarus-Schachter Theory:** Experiencing emotions is a joint effect of physiological arousal and cognitive evaluation.
- Discrete emotion approaches assume qualitatively different emotions, whereas dimensional approaches distinguish between pleasant and unpleasant states with different activation levels.

11.5 Functions of Emotions

Throughout this chapter, it was argued that emotions play an important role in the evolution of humans and other animals by preparing them to react adaptively in reoccurring, important situations. The next section will elaborate on the assumed adaptive functions of emotions.

11.5.1 Motivation and Attention

Emotions are often the driving force for action, as is evident in everyday expressions such as “I yelled at the referee because I was angry at him.” Therefore, emotions have motivational character by encouraging people to act in a certain way based on a previous experienced or imagined event or an anticipated prospect in the future. In this respect, emotions often inform the direction and intensity of behavior toward a certain goal (see the action tendencies in [Table 11.1](#)).

In addition to this motivational effect of emotions on observable behavior, emotions also influence the orientation of attention. Humans can only process a small fraction of the incoming information. Attention processes ensure that the most relevant information is processed in detail and irrelevant information is faded out. Emotions help to adjust this attentional filter to ensure that currently relevant information is processed. Thus, emotions concentrate the limited attentional resources on information that is congruent with the current emotional state of the individual. As Janelle et al. (2020) have pointed out, recent evidence suggests that the motivational intensity of an emotion may moderate the influence of affective valence (i.e., pleasantness or unpleasantness) on attentional focus. For example, unpleasant emotions high in avoidance motivation, such as anxiety and fear, tend to narrow the attentional focus and promote actions to escape threats or perceived danger. Unpleasant emotions low in avoidance motivational intensity, such as sadness, do not necessitate action and tend to broaden the attentional focus. Similar effects on attentional focus have been observed for pleasant emotions. More elaborate processing of emotion-congruent objects or events further leads to better memory recall of these object or events. Studies inside (Wilson et al., 2009) or outside of sport (e.g., Öhman et al., 2001) indicate that discrete emotions have an influence on setting of the attention filter and thus influence which information is processed elaborately and which is not.

11.5.2 Cognition

Emotions influence not only attention allocation but also numerous other cognitive processes such as learning, memory, judgment and decision-making, and creative performance (Forgas, 2006, for an overview). In general, emotions have been shown to influence the important processes of organizing and categorizing our experience. In this respect, studies have shown that an emotion or a mood is stored in memory together with the accompanying events as a global schema (Bower, 1981, 1991a, 1991b). These associative memory representations are evident in findings of mood-congruent processing and mood-dependent memory. Studies on mood-congruent processing indicate that people show a selective sensitivity to stimuli that match their current moods. For example, Gilligan and Bower (1983) have shown that stimuli that are congruent with a person's prevailing mood (e.g., images of a spider when anxious) are more likely to be noticed, attended to, and processed more elaborately. A related body of research has provided evidence for mood-dependent memory recall. People find it easier to recall certain events from memory if their current mood is congruent with their mood when they first remembered/stored an event (Eich, 1995; Eich & Macaulay, 2000; Terry et al., 2005). Studies also suggest that positive feelings lead to more efficient and creative thinking and problem-solving than negative feelings (Isen, 1987).

11.5.3 Social Functions of Emotions

A growing body of literature shows that emotions fulfill not only intrapersonal but also interpersonal functions (see Friesen et al., 2013a for a review). Interpersonal perspectives in sport provide information on the ways in which athletes' emotions and emotion regulation operate interdependently in influencing sports performance and relationships between athletes and their teammates, coaches, and parents. According to this perspective, emotions make an essential contribution to the regulation of social interactions. For example, emotions help organizing groups by binding some individuals together and distancing them from other individuals. If a person is “boiling with anger,” we would probably tend to keep our distance from that person, whereas we would probably be more likely to seek contact with a smiling person (► [Side Story: Dunbar's “Social Brain Hypothesis”](#)). Friesen et al. (2013b) illustrate how emotion regulation of others and self can operate in the context of a team environment.

Side Story

Dunbar's "Social Brain Hypothesis"

Cozolino (2006) proposed a memorable analogy to describe the nonverbal communication of emotions within "social synapse theory." Cozolino compared nonverbal communication between individuals with the neurochemical communication of nerve cells. Accordingly, people constantly send out signals into the social synapse (the space between people) which are picked up by the sensory organs of other people and transformed into electrochemical

signals that influence the thoughts and behavior of the recipients, which are again communicated over the social synapse. Cozolino (2006, p. 24) claims that the nonverbal emotional communication has an important adaptation advantage because it has allowed the organization into increasingly larger groups: "It appears that social communication has been chosen by natural selection to be of greater survival value than disguising our intentions and feelings, so much so that we even have

ways of unintentionally 'outing' ourselves to others."

In this context, Dunbar (1993) was able to demonstrate an impressive correlation between brain size and group size of living beings. This correlation has been interpreted as evidence for the coevolution of increasing computational power (i.e., larger brains) that is required to coordinate increasing complexity of life in larger groups. One aspect of this is interpreting and acting upon the emotional signals sent out by group members.

11

An influential model assigning central importance to the social function of emotions is the "EASI model" (Emotions As Social Information; van Kleef, 2009). This model originated from the social-functional perspective of emotions (Parkinson, 1996) and is based on the premise that emotions serve to communicate important social information to other people: "Emotion is not just a feeling. Emotion is for influence" (Van Kleef et al., 2011, p. 154, see also Friesen et al., 2021; Friesen et al., 2013a, 2013b for extensions to sport). The "EASI model" assumes two specific mechanisms of how emotional expressions influence observers: via (1) inferential processes and (2) affective processes. Inferential pro-

cesses describe how observers extract information from emotional signals and use it to better assess the situation and adapt their behavior accordingly (e.g., someone is angry because I was late, so I better apologize). Affective processes refer to a direct path through which observed emotional expressions influence one's own emotional reactions (e.g., when one automatically has to laugh when another person laughs; for this reason, e.g., background laughter is used in comedy series). This process is commonly referred to as **emotional contagion** in the literature (Emotional Contagion; Hatfield et al., 1993; ► [Study Box: Facial Expressions and Emotional Contagion?](#)).

Study Box

Facial Expressions and Emotional Contagion?

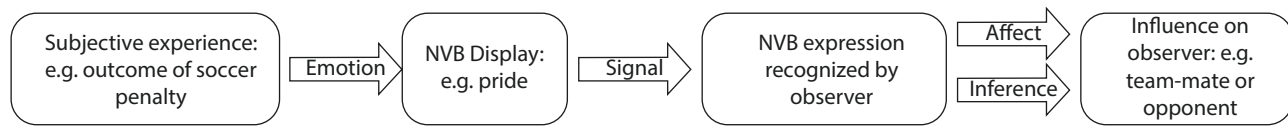
Dimberg et al. (2000) provided evidence that people tend to imitate other people's facial expressions even if they are not conscious about what they see. The participants were

shown either a happy or an angry face for only 30 ms, immediately masked by a neutral face. None of the participants could report the emotional face presented because it was not consciously perceived. Nevertheless, facial muscles (measured by facial EMG) revealed that

the participants who had been presented with a happy face showed a slightly happy expression, whereas those who had been shown a slightly angry expression reacted with a slightly more angry expression.

The relative influence of whether an observed emotional expression leads to more inferential or affective processing depends on moderating circumstances, e.g., whether the observed person is in a cooperative or competitive relationship with you. Competing relationships are more likely to be related to inferential processes, whereas cooperative relationships are more likely to

lead to emotional contagion (van Kleef, 2009). The predictions of the "EASI model" were tested and confirmed in a study in sport (► [Fig. 11.4](#); Furley et al., 2015a). This study demonstrated that the emotional expressions of pride and shame in the context of penalty shootouts have a variety of effects on opponents and teammates.



■ Fig. 11.4 Schematic presentation of the “EASI model” in sport following Furley et al. (2015a)

Sport Practice

Show Your Joy and Pride After a Successful Sporting Performance!

Moll et al. (2010) analyzed all penalties taken during World and European football championships. Not surprisingly, they were able to show that athletes expressed different emotional behavior depending whether a penalty was successful or not. More noteworthy was the finding that certain emotional expressions were related to the subsequent success or failure of the players and opponents. Players who expressed pride after a penalty kick increased the probability of their team winning the penalty shoot-out. This further increased the probability that the following competitor missed his penalty kick. More specifically, it was twice as likely that an opponent would miss his penalty if he had seen a proud emotional expression from the opposing team right before his penalty kick, compared to a player who did not celebrate his success. Although the authors of the study explain these results via the mechanism of emotional contagion, Furley et al. (2015a) pointed out that the results by Moll and colleagues are likely explained by both inferential processing (in the opponent) and emotional contagion (in teammates). An earlier study by Totterdell (2000) provided initial evidence that longer-lasting moods can also be transmitted via emotional contagion during long cricket matches and impact on a team's performance (■ Fig. 11.5).



■ Fig. 11.5 Emotional contagion in sports

11.6 Emotions and Athletic Performance

Explaining or predicting performance in sports across situations is an exceptionally difficult task. In this regard, the emotion-performance relationship seems to be particularly complicated (see Hanin, 2000; Ruiz & Robazza, 2020). Systematic investigations on the effects of discrete emotions on specific behaviors and performance in sport are rare in the literature. Nevertheless, several meta-analyses indicate a connection between sport-related performance and emotions. However, this relationship has primarily been established for anxiety or stress and athletic performance (Craft et al., 2003; Jokela & Hanin, 1999). Since ► Chap. 12 deals with anxiety and athletic performance, anxiety is not discussed in detail here. Lazarus (2000, p. 231) criticizes that stress/anxiety research and research on emotions, in general, have been conducted in relative isolation from each other:

» Stress is important in its own right, but emotion encompasses all of the important phenomena of stress. I believe that emotions provide a far richer understanding of the adaptation struggles of human and infrahuman animals.

► Functions of Emotions

- Emotions motivate to act.
- Emotions influence information processing, decisions, and behavior.
- Current theories point to important social functions of emotions in the field of communication.
- Emotions can be “contagious.”

Few studies have tested Lazarus's "cognitive-motivational theory of emotions" (1991, 2000) in sport to approach the complex relationship between discrete emotions and specific performance (where again the majority of studies are in the area of stress and anxiety). Similar to Ekman's considerations, Lazarus assumes that discrete emotions of an athlete are associated with certain "core relational themes" that are similar to the triggers and action tendencies in Ekman's Atlas of Emotions (see ■ Table 11.1). "Each emotion also entails a biologically derived action tendency or impulse that may be so strong that it is difficult to inhibit" (Lazarus, 2000, p. 243). For example, anger can be triggered by an obstacle (object or person) that prevents you from achieving a desired goal. The behavioral tendency includes increased muscle tension to overcome the obstacle. First, isolated studies could provide evidence that certain discrete emotions led to improved performance in specific sport-related tasks, e.g., anger led to increased strength when lifting, holding weights, or throwing (Rathsclag & Memmert, 2013; Woodman et al., 2009).

? What does IZOF mean?

- Emotion-performance relationships in sport are very complex. Simple models are likely inadequate (e.g., inverted-U hypothesis, ► Chap. 12).
- The influence of emotions on athletic performance can only be understood as an individualized person-environment interaction.
- Person-environment interactions describe the relationship between task characteristics and the resources of a person.
- Research on emotions in sport should relate the optimal and suboptimal emotional experiences of an athlete to their successful and unsuccessful performances.
- IZOF envisages five dimensions of emotions: form, content, intensity, time, and context.
- Athletes should use their own labels to describe emotions before, during, and after competition since common measurement instruments do not include about 80% of the idiosyncratic emotion labels used by athletes (Ruiz & Hanin, 2004; Syrjä & Hanin, 1998).

Self-Reflection

Remember personal emotional experiences in sports. Try to think of as many different emotional experiences (joy, pride, shame, anger, relief, etc.) as possible and answer the following questions as accurately as possible:

- How would you describe the trigger of the emotional experience?
- Was this experience linked to any action tendencies or thought processes? If so, to which ones?
- What else do you associate this emotional experience with?

The emotion-performance relationship in sport is moderated by interindividual appraisal styles and **coping** strategies of athletes (Hanin, 2007). For this reason, the first textbook on emotions in sport (Hanin, 2000) postulates a highly individualized approach (individual zones of optimal functioning, IZOF) for understanding emotions in sport. Twenty years later, a second textbook has been published (Ruiz & Robazza, 2021) summarizing contemporary knowledge on a wide range of emotion and emotion-related feeling states in sport through a multidisciplinary perspective. Although largely expanded and integrated, Hanin's emphasis and contribution on individual performance states, experiences, and emotion self-regulation strategies are still valid today.

11.7 Emotion Regulation

Emotion Regulation

Emotion regulation describes the process of how people influence what emotions are experienced, when they occur, and how they are experienced and expressed (Gross, 1998).

Emotion regulation can be distinguished from the term "coping," as coping is always related to a stressor and can be problem-focused or emotion-focused. Emotion regulation differs in that people regulate to a standard and, therefore, engage in regulation strategies when they perceived a discrepancy between current and desired emotions. Emotion regulation is not always stress related; people can engage in emotion regulation strategies to remain happy or try to feel happier, for example. A dominant model of emotion regulation (Gross, 1998, 2014) is the process model. In this model, there are five families of emotion regulation: (a) situation selection, (b) situation modification, (c) attention orientation, (d) modification of cognition, and (e) suppression of emotions (see Lane et al., 2012a, for a review of emotion regulation in sport).

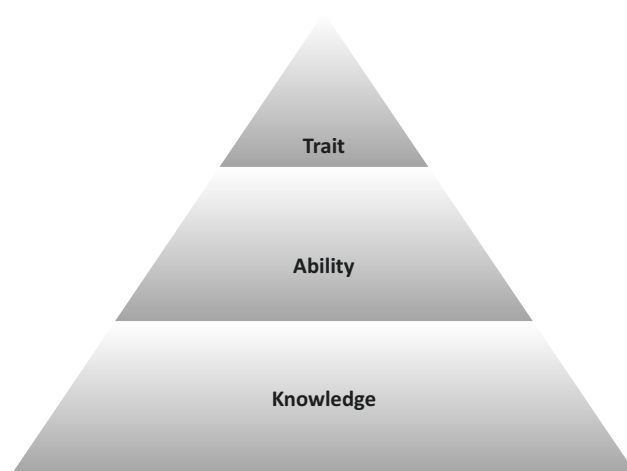
By seeking out or avoiding certain situations, (a) people can influence their emotional experience. Athletes who feel nervous and possibly ashamed dur-

ing public television interviews can influence their emotional experience, for example, by avoiding giving television interviews. However, they could also modify this situation (b) by not giving an interview immediately after a match and waiting until they have calmed down and rested. Through attention orientation (c), athletes can control which aspects of a situation they focus on. For example, if basketball players focus on the situation in which they felt they were being treated unfairly by the referee, this will have different emotional consequences than if they focus their attention on their own performance. Analogous to attention orientation, athletes can modify the cognitive assessment of a situation (d). If a handball penalty is assessed as a threat due to the chance of letting one's team down, this will probably have different emotional consequences than if the situation is assessed as a challenge in helping one's own team. All these examples illustrate how the emotional experience can be regulated at different points in time and thus lead to different emotional reactions (experience and physiological reactions) and associated behavioral tendencies. The last point of emotion suppression (e) describes how athletes can suppress emotions to regulate emotions. To return to the example of the football penalty kick, penalty shooters often show the behavioral tendency to get out of a penalty situation as quickly as possible in stressful conditions and do not take their time to prepare for the penalty kick (Furley et al., 2012b). When there is not a clear opportunity to change or modify the situation, alter attention, or think differently about the situation, and knowing this can lead to negative performance (Jordet & Hartman, 2008), penalty shooters can instruct themselves to take two deep breaths after the referee's whistle before initiating the run-up.

The different points in the emotional regulation process can be more or less automatic or controlled as well as nonconscious or conscious. Through learning processes or sport psychological interventions, conscious, controlled emotional regulation processes can be trained and become increasingly autonomous (e.g., Furley et al., 2015b). In this way, the emotional experience of a person can be intentionally changed and stabilized.

11.8 Emotional Intelligence

In the previous sections, emotions have been treated as transient states. However, it is also possible to investigate the stable aspects of an individual's emotional experience (Lazarus, 2000). The most prominent concept reflecting these relatively stable emotional characteristics has been termed **emotional intelligence (EI)**



■ Fig. 11.6 Illustration of the “three-part model of emotional intelligence” (Nelis et al., 2009)

(Petrides & Furnham, 2003; Salovey & Mayer, 1990). Since the emergence of the concept in the 1990s, there has been much debate on EI, especially regarding the conception of EI as a trait or as an ability. The “tripartite model of EI” (■ Fig. 11.6) embraced the trait and ability models by assuming there are different components of EI that perform important functions: (1) a knowledge component, (2) an ability component, and, (3) trait component (Nelis et al., 2009).

Emotional Intelligence

Emotional intelligence describes the relatively stable aspects of an individual's emotional experience and actions. It is divided into knowledge, ability, and trait components. Emotional intelligence is relatively stable but can be improved via training (Laborde et al., 2021).

The **knowledge level** refers to the complexity and range of emotional knowledge. It describes the emotional knowledge of a person and how this knowledge is used to deal with emotionally charged situations.

The **ability level** refers to applying emotional knowledge in an emotional situation and using strategies for how to act. The focus here is not on what people know, but on how they can act.

The **trait level** refers to emotional dispositions, namely, the tendency to behave in a certain way in emotional situations. The focus here is not on what people know or can do, but on what they habitually do.

These three levels of EI are loosely connected: “Knowledge is not always immediately transformed into abilities, which in turn are not always put into practice” (Nelis et al., 2009, p. 36).

11.8.1 Emotional Intelligence in Sport

A review paper on the influence of EI on athletic performance (Laborde et al., 2016a) showed a positive correlation between EI and athletic performance. Studies suggest that the ability component of EI has a positive effect on objective performance criteria over an entire season in cricket (Crombie et al., 2009), basketball (Zizzi et al., 2003), and ice hockey (Perlini & Halverson, 2006). Emotional intelligence has been found to relate to emotional states experienced before competition, which relate to performance (Lane et al., 2012b; Lane & Wilson, 2011). Furthermore, EI has been suggested to influence subjective performance happiness. In this context, stress perception, coping strategies, and the assessment of situations as challenging rather than threatening were identified as mechanisms (Laborde et al., 2014a). Moreover, EI is found to make people more resistant to pressure and severe stress. Athletes with a high trait EI were able to perform better under pressure by using more efficient coping strategies (Laborde et al., 2012) and were found to have superior physiological stress resistance (Laborde et al., 2011, 2014b, 2015b).

EI is important not only for athletes but also for coaches (Hopkins et al., 2011; Thelwell et al., 2008). In general, coaches with higher EI have been shown to coach more effectively, for example, by exhibiting a more efficient management style and by paying more attention to athletes' needs.

With regard to physical activity and exercise, a positive relationship between trait EI and an active lifestyle has been documented, which has significant benefits for both **health and well-being**. Trait EI is believed to promote an active lifestyle through motivational mechanisms that play a role in initiating and maintaining physical activity behavior (Saklofske et al., 2007a,b; Solanki & Lane, 2010). Consequently, trait EI goes hand in hand with an active and healthy personality (Laborde et al., 2017).

11.8.2 Training of EI in and Through Sport

EI training has already been implemented in a variety of sports including cricket (Crombie et al., 2011), netball (Barlow & Banks, 2014), and rugby (Campo et al., 2016). An overview article on EI training (Campo et al., 2015) recommends that successful EI training should ideally be based on the tripartite model of EI (Nelis et al., 2009). The organization of such training

aims to increase both the emotional knowledge and ability component to initiate changes at the trait level of EI. Successful EI training programs have demonstrated positive long-term consequences on the psychological and physiological level (Kotsou et al., 2011; Nelis et al., 2011).

➤ Emotional Intelligence

- EI describes how a person deals with emotions in different situations.
- EI is subdivided into a knowledge, ability, and trait level.
- EI has a positive influence on athletes and coaches in sports and exercise.
- EI can be trained.

11.9 Measurement of Emotions and Emotional Intelligence

11.9.1 Measurement of Emotions

The controversies in emotion research addressed in the introduction are particularly evident in the scientific investigation and measurement of emotions. As a result, no standard paradigm for the investigation and measurement of emotions has been established so far.

❓ Why is the scientific study of emotions a real challenge?

A quotation from Alan Fridlund vividly illustrates the problem (Beck, 2015): “At the heart of ‘emotion’ is the experience of emotion, and this can’t be measured. Recorded, maybe, but not measured. This leaves scientists studying ‘emotion’ trying instead to measure everything around it.”

- Emotions are multidimensional **constructs**.
- Exploring them requires a **multi-methodological approach**.
- Inducing emotions under standardized conditions is a challenge.
- Intense emotional states are ethically hardly feasible in the laboratory.

11.9.1.1 Inducing Emotions

Different approaches have been used in emotion research to induce and investigate emotional states (Table 11.2; Coan & Allen, 2007; Quigley et al., 2014; for overviews; Wetherell et al., 2018, for an overview of techniques of inducing stress).

Table 11.2 Selected emotion induction techniques (based on Quigley et al., 2014)

Method	Stimulus material and references	Advantages	Disadvantages
Images	Bradley et al. (2001) International Affective Picture System (IAPS) (Lang et al., 2008)	Easy presentation Economical	Do not include all emotions or are not suitable for all
Videos	Gross and Levenson (1995) Philippot (1993) Schaefer et al. (2010)	Easy presentation Economical More emotional information compared to pictures	Familiarity of the content can cause variability
Faces	Ekman and Friesen (1978) Matsumoto and Ekman (1988) Beaupré and Hess (2005)	Easy presentation Economical Faces have high emotional valence	Not clear whether faces trigger emotions or activate concepts
Sounds/voices	IADS (Bradley & Lang, 2007) Bliss-Moreau et al. (2010) Simon-Thomas et al. (2009)	Easy presentation Economical	Not well researched whether emotions are actually induced
Music	Eich and Metcalfe (1989)	Easy presentation Can preserve emotional states over a long period	Not suitable to evoke discrete emotions (rather valence)
Words	Bradley and Lang (1999) Stevenson et al. (2007)	Simple presentation Relatively low effect strength	Not clear whether words trigger emotions or activate concepts
Imagination techniques and recall	Wilson-Mendenhall et al. (2011) Olatunji et al. (2009) Velten (1968) Carter et al. (2002)	External validity Can be ideographically manipulated	Great interindividual variability in ability of imagination
Movement and posture: facial (body) feedback paradigm	Strack et al. (1988) Duclos et al. (1989) Stepper and Strack (1993)	Relatively implicit method to induce emotions	Needs good cover story to avoid demand effects
Staged interactions	Cohen et al. (1996) DeSteno et al. (2006)	External validity	Complicated Needs good cover story and actors
In situ observation and motivated performance tasks	TSST (Kirschbaum et al., 1993) Akinola and Mendes (2008) Teachman (2007) Castanier et al. (2011)	External validity	Complicated Expensive Ethical concerns
Virtual reality	Riva et al. (2007)	External validity in combination with experimental control	Complicated Expensive
Physiological and pharmacological manipulation	z.B. epinephrine-injections (Schachter & Singer, 1962) Oxytocin injections (Norman et al., 2011) Sport (Ekkekakis et al., 2011)	High impact	Medical expertise needed Expensive Ethical concerns Needs good cover story

This list represents measures from the entire scientific spectrum. The sports reference is made in the measurement section

? How can emotions be induced and investigated?

- Presentation of standardized stimuli material with validated emotional content (pictures, films, faces, sounds/voices, music, words)
- Imagination techniques and memories
- Movement and posture: facial (bodily) feedback paradigm
- Staged interactions
- In situ observation and motivated performance tasks
- Virtual reality
- Physiological (e.g., sports) and pharmacological (e.g., drugs) manipulation

11.9.1.2 Measuring Induced States

Depending on the author, there are substantial differences in how many dimensions should be distinguished in the measurement of emotions. According to Scherer (2005), there are five dimensions of emotions: cognitive, neuropsychological, motivational, motor-expressive, and subjective emotional component (■ Fig. 11.7).

It is important to note that the various components of emotions can function separately from one another (Scherer, 2005). It is therefore advisable to measure several components at the same time (Mauss & Robinson, 2009). It should be emphasized, however, that the assessment of all the different components is difficult to realize and therefore this should not be made a standard procedure. We aim to inform sports scientists and sport psychologists about the possibilities of measuring different emotional components so that they can make an informed choice about appropriate measurement tools to answer their research question.

■ Cognitive Component

The cognitive component refers to the appraisal of objects, events, and situations. To evaluate the cognitive assessment component, a contemporary questionnaire in sport, the “precompetitive appraisal measure” (PAM; Wolf et al., 2015), was developed. The PAM is based on the “cognitive-motivational-relational theory” (Lazarus, 1999) and records the primary (How do I assess the situ-

ation?) and secondary assessment processes (How do I assess my resources to cope with the situation?) of a person. It aims to determine what generates precompetitive emotions.

■ Neurophysiological Component

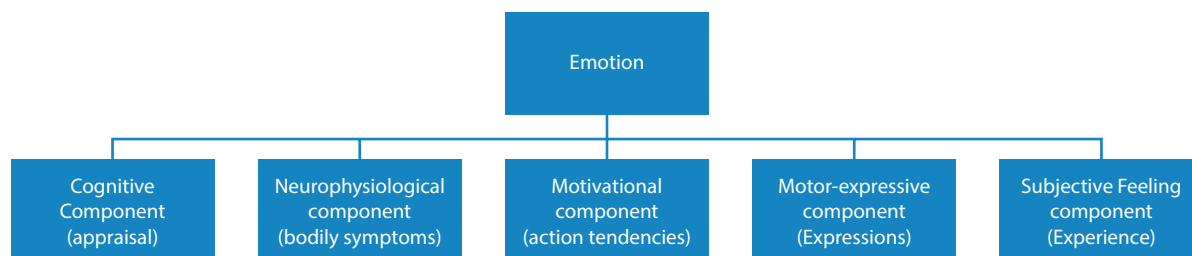
The most important neurophysiological parameters related to emotions and athletic performance are electrodermal activity, heart rate, heart rate variability, hormone concentrations, and brain measurements. The collection of neurophysiological parameters in sport can be problematic, since these parameters change not only in relation to emotions but also in relation to movement and physical activity.

■ ■ Electrodermal Activity

The electrodermal activity of the skin coincides with affective, motivational, and attention processes of the central nervous system (LaCaille et al., 2013). The electrodermal activity of the skin can be divided into a tonic and a phasic category.

The tonic value of skin resistance (also known as conductance) is the absolute degree of resistance or conductance at a given moment in the absence of a measurable phasic reaction. This value is known as skin conductance. In tonic measurements, rest conditions prevail, whereas in phasic measurements, certain stimuli are used to induce activation in the subject (Dawson et al., 2000).

Electrodermal activity is used as an indicator of arousal (intensity), but not of the valence of the emotion. The skin conductance response correlates positively with emotional arousal regardless of valence (Bradley & Lang, 2000). It is assumed that the skin conductance response is regulated exclusively by the sympathetic nervous system (Venables & Christie, 1980). The sympathetic innervations of the sweat glands induce changes in skin conductivity (Gutrecht, 1994), which is associated with emotional arousal in a variety of situations (Dawson et al., 2000). In sports, skin conductance response has rarely been used as a measure of emotion, as it is strongly associated with physical activity (for exceptions, see Collet et al., 2003; Rada et al., 1995).



■ Fig. 11.7 The five relevant components for measuring emotions according to Scherer (2005)

■ ■ Heart Rate

To a certain extent, **heart rate** can indicate the valence and intensity of an emotion. Bradley and Lang (2000) demonstrated that emotional images led to systematic changes in the classic three-phase pattern—an initial delay, an acceleration component, and a secondary delay—of heart rate. More specifically, it was shown that affective valence contributed to the extent of initial deceleration and acceleration activity, with unpleasant stimuli producing a stronger initial deceleration and pleasant stimuli producing a stronger peak acceleration.

There are however many other variables that influence heart rate, such as posture, breathing, and physical constitution. These variables may obscure an affective influence on heart rate (Bradley & Lang, 2000). Nevertheless, it might be possible to measure the effect of affective valence on heart rate when the processing context is appropriately controlled. Although physical activity complicates the interpretation of heart rate as an emotional indicator in sport, some studies in sport have demonstrated association between heart rate and anxiety in climbing (Oudejans & Pijpers, 2010) and gymnastics (Tremayne & Barry, 1988).

■ ■ Heart Rate Variability

Heart rate variability (HRV) can be calculated from the time intervals between the R-R peaks of the ECG. HRV is never constant but varies from beat to beat and corresponds to the variability of RR intervals (i.e., intervals between successive R-R peaks; Niskanen et al., 2004).

HRV reflects the interaction between the sympathetic and parasympathetic branches of the autonomic nervous system (Malik, 1996). Both sympathetic and parasympathetic branches of the autonomic nervous system are involved in emotions (Levenson, 2003); therefore, HRV can be considered an objective measure of emotional response (Appelhans & Luecken, 2006). Of the two branches, HRV can only indicate the activity of the parasympathetic nervous system regulating cardiac functioning, also known as cardiac vagal activity (Malik, 1996). This is of particular interest for the sports domain since it has been postulated that resting cardiac vagal activity has a positive relationship to cognitive functions, emotion regulation, and health according to the “neurovisceral integration model” (Thayer et al., 2009).

In sport, athletes who show a lower decrease in cardiac vagal activity under pressure show better decision-making performance (Laborde & Raab, 2013; Laborde et al., 2014c) and memory performance (Laborde et al., 2015a), which is consistent with the “neurovisceral integration model” (Thayer et al., 2009). Emotional intelligence has also been associated with higher resting

cardiac vagal activity and lower cardiac vagal activity reduction under pressure (Laborde et al., 2011, 2015b), which is indicative of a biological basis of more adaptive emotional dispositions.

■ ■ Hormonal Measurement

Measurement of **hormones**, in particular cortisol, can serve as indicators of anxiety or stress (Denson et al., 2009a, 2009b). **Cortisol** measurement in saliva is particularly helpful because it is noninvasive, painless, and thus ethically justifiable. It can be carried out without medical expertise (Kirschbaum & Hellhammer, 2000).

In athletic competition, an increase in cortisol levels often coincides with an individual’s stress experience (Filaire et al., 1996, 1999, 2001). Research has, for example, shown that the closer a competitive encounter approaches, the higher the cortisol level is (Strahler et al., 2010). The cortisol response of athletes to stressful encounters has been shown to be less important at the highest-performance level compared to low-performance levels (Moya-Albiol et al., 2001). However, a growing number of studies suggests that there is a negative correlation between cortisol and athletic performance (Lautenbach et al., 2014, 2015a, 2015b).

■ ■ Measurement of Brain Activity

Measurements of brain activity are a promising new development in the field of emotion and performance (see Bertollo et al., 2020, for a recent overview). In sports, for example, **functional magnetic resonance imaging (fMRI)** has been used to investigate the effect of cognitive interventions on coping with sports failure at the neurophysiological level (Davis et al., 2008). A further study combined fMRI with neuroendocrine measurements to investigate the correlates of neuronal activation with cortisol and testosterone (Davis et al., 2012). A challenge for researchers is adapting athletic emotional tasks to the particularities of fMRI (Daamen & Raab, 2012) since a large number of observations is required to derive reliable brain activation patterns.

Another method for measuring brain activity with a higher time resolution is **electroencephalography (EEG)**. Although electroencephalography has been extensively used in sports to study attention processes (e.g., Sanchez-Lopez et al., 2016), this method has rarely been used to study emotional processes in sports and needs to be further established (for an exception, see Ring et al., 2016).

First studies suggest that neurofeedback may be a useful applied tool to improve performance under pressure (Ring et al., 2014). The development of mobile EEG methods might provide a promising perspective to improve the understanding of emotions in sports (Park et al., 2015).

■ Motivational Component

The motivational component within the measurement of emotions reflects the action tendencies associated with discrete emotions (Ekman & Cordaro, 2011; Lazarus, 2000). The motivational component can affect behavior by interrupting or inhibiting current behavior and generating new goals and plans (Scherer, 2005). The motivational component of emotions is based on circuits in the brain that respond to appetitive and aversive stimuli (Lang & Bradley, 2010). Most commonly the motivational component is assessed via self-report or behavioral observation. Another possibility to investigate the motivational component of emotions is the

use of EEG and event-related potentials. This method has been used, for example, to demonstrate the influence of emotions on risk preference and outcome assessment during a decision task (Zhao et al., 2016).

■ Motor-Expressive Component

According to Bradley and Lang (2000), behavioral events can be assessed either by direct actions (e.g., emotional expressions, approach, avoidance, flight, attack, defensive reflexes) or indirect indicators (e.g., response latency, amplitude; ► **Study Box: Winning Athletes Show Discrete Emotional Expression After a Triumph**).

Study Box

Winning Athletes Show Discrete Emotional Expression After a Triumph

Matsumoto and Hwang (2012) recorded the spontaneous reactions of successful judoka at the 2004 Olympics. Detailed evaluation of the nonverbal behaviors of the athletes showed that the emotional expressions after success or failure were dif-

ferent from known emotional expressions like pride. This was confirmed in a second study where trained coders evaluated the behavior. The authors argued that certain expressions of superiority and triumph have developed over the course of evolution (► Fig. 11.8), which help to establish status and dominance in

social groups and thus establish hierarchies. This line of argumentation was further confirmed by the finding that the expressions of superiority and triumph were also shown by athletes at the Paralympics who were blind from birth and therefore could not have learned these expressions (Tracy & Matsumoto, 2008).



► Fig. 11.8 Athletes show a discrete emotional expression after a success

The effects of emotions on muscle activity and movement behavior can be evaluated by controlled monitoring of changes in movement patterns and muscle tension (Pijpers et al., 2003, 2005). These studies show that it is possible to use electromyographic data to assess qualitative differences in muscle tension and movement patterns as a function of emotional state. It has also been shown that emotions influence the speed and initiation time of targeted movements (Beatty et al., 2016).

The influence of emotions on physical performance was demonstrated in various sport-related tasks, such as the strength of the finger muscles, jump height of a countermovement jump, the speed of a thrown ball (Rathschlag & Memmert, 2013), and sprint performance (Rathschlag & Memmert, 2014; ■ Fig. 11.9). Furthermore, it has been demonstrated that changes in performance under pressure are mediated by emotionally induced changes in muscle activity and kinematic parameters (Cooke et al., 2010; ► Side Story: What Do Movements Say About Emotions in Sport?).



■ Fig. 11.9 Emotions can influence sprint performance, for example (Rathschlag & Memmert, 2014)

Side Story

What Do Movements Say About Emotions in Sport?

Motion recordings can make an important contribution to the analysis of the motor component of emotions in sports, even though there is a controversial discussion about whether it is possible to reliably draw conclusions about emotional states based on movement (Castellano et al., 2007). First studies show that it is possible to identify discrete emotions through movement (Li et al., 2016; Vernazza-Martin et al., 2015, ■ Fig. 11.10).



■ Fig. 11.10 Expression of emotions in sports

■ Subjective Emotional Component

There are two main methods for obtaining self-evaluations of emotional experiences: the discrete emotion approach and the dimensions approach (Scherer, 2005).

■ ■ Discrete Emotions

There are two approaches for measuring discrete emotions: individualized (**idiosyncratic**) and group-oriented (**nomothetic**).

Idiosyncratic The individualized approach was initiated by the work of Hanin and his colleagues with the IZOF (Hanin, 2000; Jokela & Hanin, 1999). An individualized approach allows scientists to get closer to the experienced emotions of athletes and to grasp their idiosyncratic nature for each individual athlete. To allow for group analysis of IZOF data, a method has been proposed that combines performance-enhancing and performance-reducing emotions (Laborde et al., 2016b). The IZOF was further developed and combined with physiological measurements to allow predictions of when athletes are in an optimal emotional performance range (Bertollo et al., 2012). Since theory testing and the synthesis of data from different studies is difficult with this idiosyncratic approach (Jones et al., 2005), researchers usually choose the group-oriented (nomothetic) approach.

Nomothetic For the group-oriented approach, there are several standardized sport-specific measures that focus on one specific emotion, e.g., fear. These, for example, include the “Competitive State Anxiety Inventory-2” (Martens et al., 1990b), which has been revised to an updated version (Martinent et al., 2010), and the “Sport Anxiety Scale-2” (Smith et al., 2006).

Non-sport-specific scales are often used to assess a broader range of affective states: the Profile of Mood States (POMS; McNair et al., 1971; although it refers to moods rather than emotions), the Brunel Mood Scale (BRUMS; Terry et al., 2003), and the Positive Affect and Negative Affect Schedule (PANAS; Watson et al., 1988). The POMS and BRUMS assess anger, tension, depression, fatigue, confusion, tension, and vigour.

Even though the POMS and the PANAS have been used in athletic contexts, they were not designed to assess emotions in sport. Therefore, the sport-specific questionnaire “Sport Emotion Questionnaire” (SEQ; Jones et al., 2005) was developed. It contains 22 emotion adjectives from five different dimensions (anger, anxiety, dejection, excitement, and happiness) and has been validated to measure emotions before a competition. The SEQ correlates very closely to the BRUMS,

and the resultant measure was very similar to the model developed initially for clinical use. As emotions are universal, one might wonder why specific emotions might arise in certain contexts, and rather than seek different emotions, it is more reasonable to expect the intensity of emotions to vary by context. Thus we expect anxiety to be high before competition and low after competition. Further instruments measuring emotion-related feelings in sport (“Psychobiosocial States Scale, Trait Version”, PBS-ST, and Psychobiosocial Experience Semantic Differential scale in sport, PESD-Sport) were developed by Robazza et al. (2016, 2021; see Ruiz et al., 2019, for a state version of the questionnaire).

The Competitive Aggressiveness and Anger Scale (CAAS; Maxwell & Moores, 2007) was used to reveal correlations between certain emotions and illegal aggressive behavior in rugby (Maxwell & Visek, 2009). In addition, picture tests have been used to measure people’s attitudes toward participating in physically and emotionally challenging tasks in physical education (Bortoli & Robazza, 1994) and high-risk sports (Robazza et al., 2006; ■ Fig. 11.11).

■ ■ Dimensional Approach

This category of measuring tools typically assesses different dimensions of emotions via pencil and paper scales. Common dimensions that are measured are the **hedonic valence** (pleasant vs. unpleasant) and the activation level (i.e., arousal). A frequently used instrument is the “Affect Grid” (Russell et al., 1989) which evaluates **hedonic valence** and **intensity**. The sport-related “Affect Grid” is based on the same principle (Woodman et al., 2009). It evaluates two independent dimensions of affect: activation and valence (pleasantness). It is presented as a 9 × 9 grid (■ Fig. 11.12): The vertical axis evaluates activation (from extremely low to extremely high), and the horizontal axis evaluates hedonic valence (from unpleasant feeling to pleasant feeling).

Another popular instrument is the “activation-deactivation checklist” (Thayer, 1989). It records the activation dimension and was used to assess affective sensation after physical activity (Ekkekakis et al., 2005). The “Affect Scale” (Hardy & Rejeski, 1989) is a bipolar evaluation scale used to assess emotional reactions during training.

The “Visual Analog Scale” is a biaxial orthogonal grid (200 mm axes with the poles “not at all” and “very much”) that has been used to measure the arousal and hedonic valence dimensions. The scale is frequently used as a quick and effective measure to control emotional manipulations (Davis et al., 2010).



■ **Fig. 11.11** In big wave surfing, the athletes expose themselves to an enormous risk

■ **Fig. 11.12** “Sport Affect Grid.” Participants are asked to mark a section of the grid with an X that represents the current emotional state. The score for activation and the hedonic valence of feelings are calculated separately by converting the position of the X on each axis to a value from 1 to 9 (Woodman et al., 2009)

Extremely High Intensity Extremely Low									
Unpleasant Feeling Pleasant Feeling Feeling									

In general, dimensional measures provide less information than inventories assessing discrete emotions but can be helpful for quick affective assessments.

➤ Measuring Emotions

Measuring emotions is complex. The following indicators provide information about a person's emotional states:

- Cognitive component
- Neuropsychological component
- Motivational component
- Motor-expressive component
- Subjective emotional component

11.9.2 Measuring Emotional Intelligence

Emotional intelligence as a **personality trait** is usually measured with personality questionnaires, while EI as

an ability is measured with emotional intelligence tests (Petrides, 2009a). Regarding the measurement of EI as a personality trait, nine different questionnaires have been used in the sport context (Laborde et al., 2016a). A full description of all these questionnaires is beyond the scope of this chapter, so we will limit ourselves here to the description of the emotional intelligence questionnaire (Petrides, 2009b).

This questionnaire shows the best psychometric properties (Petrides, 2009a) and the highest external validity. Furthermore, the questionnaire showed associations with physiological and hormonal responses (Laborde et al., 2011, 2014a, 2014b, 2014c, 2015a, 2015b). The questionnaire contains 153 terms that can be used to calculate 15 subscales included in 4 factors (i.e., well-being, self-control, emotionality, and sociability) and a global EI score. This factor structure has been replicated using a sample of sportsmen and women (Laborde et al. 2014a, 2014b, 2014c).

In sport, Lane et al. (2009) examined the validity of the frequently used emotional intelligence scale (Schutte et al., 1998). Lane, Meyer et al. (2009) found that many items showed poor face validity, and Davies et al. (2010) found evidence to support a brief ten-item version. For researchers wishing to obtain a brief measure of emotional intelligence, for example, in contexts where brevity is important, the short measure developed by Davies et al. (2010) has many desirable qualities.

Regarding the measurement of EI as an ability, tests like the “Mayer-Salovey-Caruso Emotional Intelligence Test” (MSCEIT; Mayer et al., 2002), “Test of Emotional Intelligence” (TEMINT; Blickle et al., 2011), and “Ability Emotional Intelligence Measure” (AEIM; Warwick et al., 2010) have been developed (Warwick et al., 2010).

The MSCEIT is the most commonly used ability EI test and the only one used in sport (Crombie et al., 2009, 2011). It evaluates four emotional dimensions: the ability (1) to perceive emotions, (2) to use emotions to facilitate thoughts, (3) to understand emotions, and (4) to control emotions. The four dimensions are assessed with 141 items. Although this measure has been criticized (Fiori et al., 2014), it currently represents the most widely used measure in the field of sport.

Learning Control Questions

Basic:

1. What is an emotion? How does an emotion differ from a mood?
2. What is the main statement of Darwin’s “theory of emotional expression” and why are emotional expressions central to understanding emotions?
3. What are the findings on Darwin’s theoretical assumptions in sport? Describe them and discuss their significance for sport practice.
4. What do the studies dealing with emotional expressions say about the influence of nature and nurture in the field of emotions?
5. What does facial feedback mean? Discuss the significance of these findings for emotional experience and behavior.

Advanced:

6. Describe Ekman’s “neurocultural theory of emotions” in your own words.
7. What is meant by characterizing emotions as universal?
8. What emotions are described as universal/basic? Describe their assumed function, states, triggers, behavioral tendencies, and associated moods.

9. Describe the most important anatomical structures and the associated physiological processes related to emotions.
10. How are emotional physiological processes related to information processing, decision-making, and behavior?
11. Name the three theoretical approaches to describe the connection between physiology and emotional experience.
12. How does the discrete emotion approach differ from the dimensional approach? What are the arguments for the discrete emotion approach?
13. What are the functions of emotions?
14. Which findings support the interpersonal functions of emotions?

Experts:

15. Describe the “EASI model” and discuss the significance of the model for sports.
16. What is the relationship between emotions and sports performance? Outline key assumptions and findings.
17. What are the five aspects that can be addressed by emotion regulation?
18. What is emotional intelligence? What is the relationship between emotional intelligence and sports performance?
19. List different ways to induce emotions.
20. Which dimensions can be distinguished when measuring emotions in sport?

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